

Course One

Foundations of Data Science



Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- Complete the questions in the Course 1 PACE strategy document
- Answer the questions in the Jupyter notebook project file
- Complete coding prep work on project's Jupyter notebook
- Summarize the column Dtypes
- Communicate important findings in the form of an executive summary

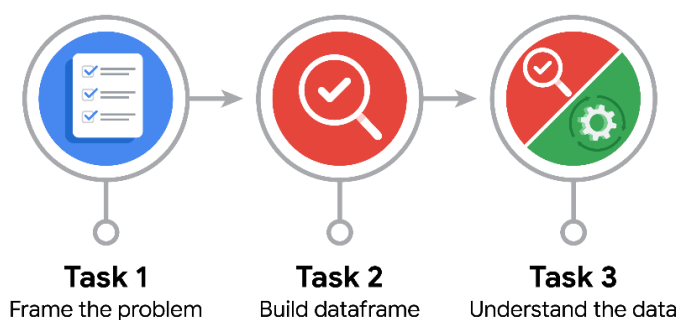
Relevant Interview Questions

Completing the end-of-course project will help you respond to these types of questions that are often asked during the interview process:

- Describe the steps you would take to clean and transform an unstructured data set.
- What specific things might you look for as part of your cleaning process?
- What are some of the outliers, anomalies, or unusual things you might look for in the data cleaning process that might impact analyses or ability to create insights?

Reference Guide

This project has three tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations



PACE: Plan Stage

- How can you best prepare to understand and organize the information provided?

Reviewing the project objectives, meeting notes, and dataset documentation, Then organize the data into a pandas Data Frame and identify the variables needed for future analysis.

- What follow-along and self-review codebooks will help you perform this work?

The panda's documents, Python material, course notes, NYC Taxi dataset, and Notebook documents.

- What are some additional activities a resourceful learner would perform before starting to code?

Review the dataset documentation to understand and verify that python libraries are installed. Outline the workflow.



PACE: Analyze Stage

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

Yes, the dataset contains trip information such as pickup and dropoff times, trip distances, passenger count, payment type, fare costs, tips, tolls, and total fare.

- How would you build summary data frame statistics and assess the main and max range of the data?

Use panda's functions such as `info()`, `describe()`, etc. inspects for any updates dupes and looks through the dataset to view and minimize anomalies.

- Do the averages of any of the data variables look unusual? Can you describe the interval data?

This will be determined after reviewing the summary stats for the notebook.



PACE: Construct Stage

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.



PACE: Execute Stage

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing exploration data analysis?

Verify the quality of the dataset by checking for missing values, dupes, incorrect data types, unrealistic values.

- What data initially presents as containing anomalies?

Possible anomalies include missing values, dupes, negatives amounts, incorrect passenger counts, etc.

- What additional types of data could strengthen this dataset?

Weather conditions, traffic info, holidays, specials events, construction site.